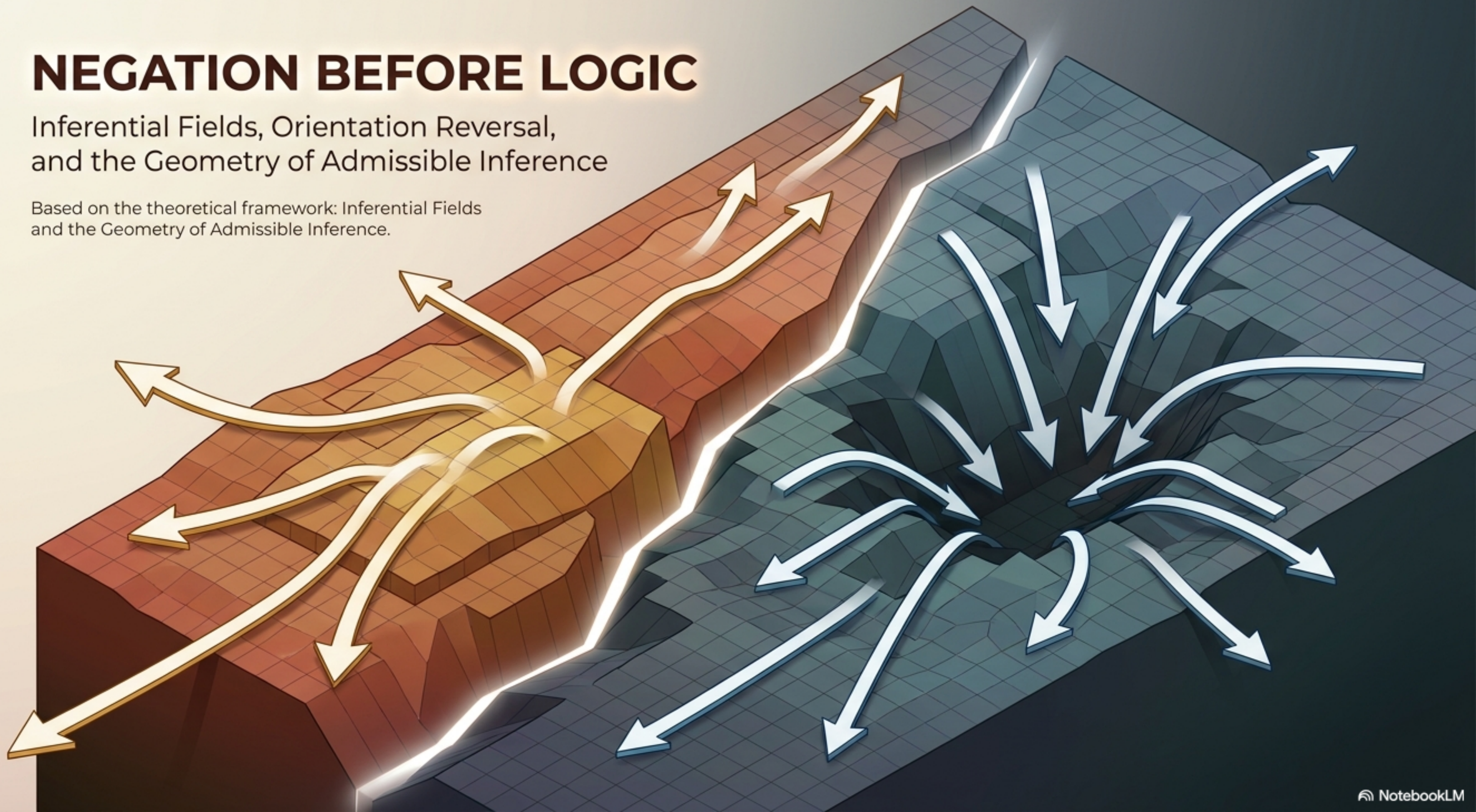


NEGATION BEFORE LOGIC

Inferential Fields, Orientation Reversal,
and the Geometry of Admissible Inference

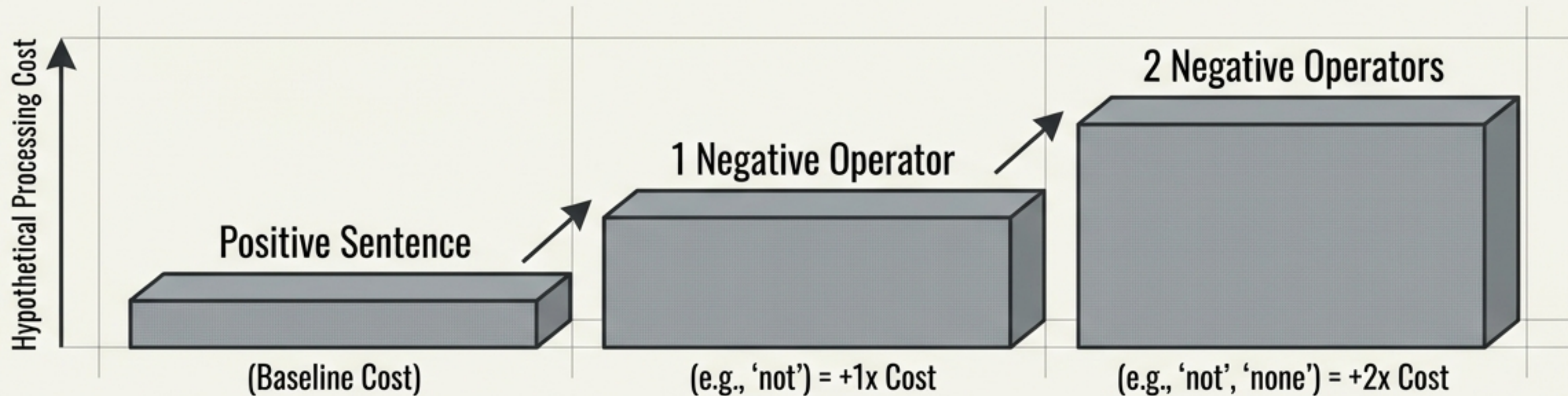
Based on the theoretical framework: Inferential Fields
and the Geometry of Admissible Inference.



The Intuitive (But Flawed) Paradigm: The Operator-Count Hypothesis

The dominant theory of negation processing is mathematically natural: processing cost should increase monotonically with the number of negative expressions encountered during parsing.

One negation is hard. Two negations are harder.



"This hypothesis has the virtue of simplicity and the vice of being wrong in precisely the cases that matter most theoretically."

The Systemic Failure of Operator-Counting

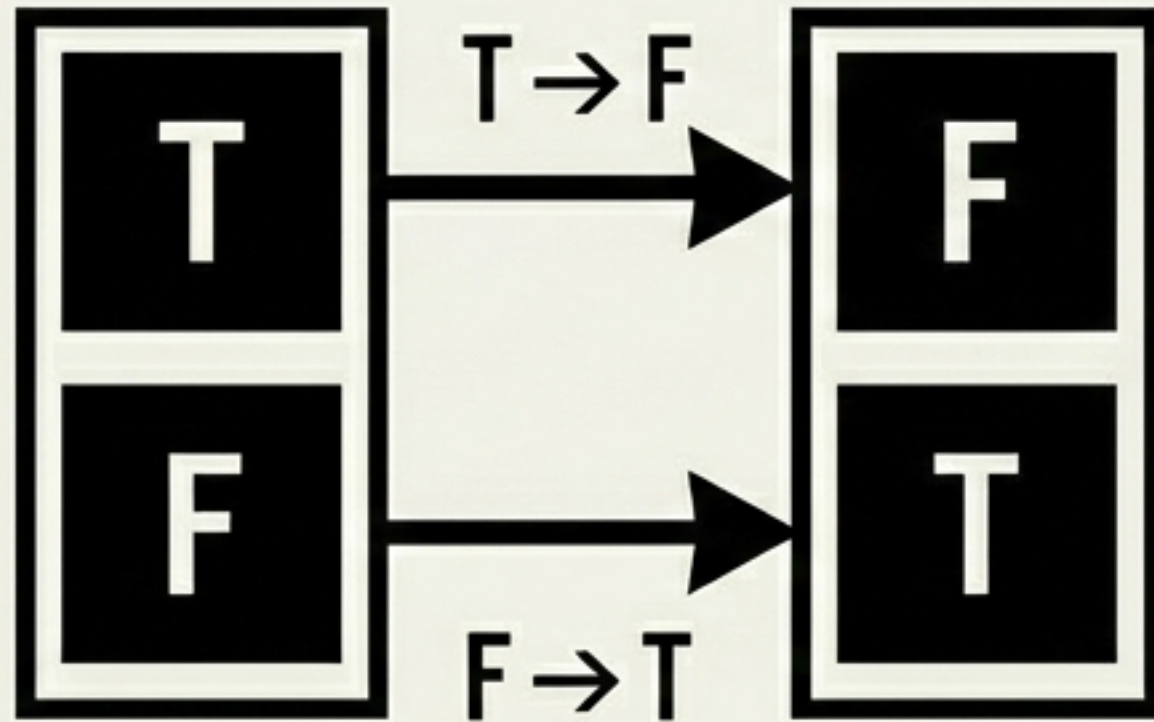
The experimental record does not show random noise; it shows systematic deviation pointing toward an entirely different underlying variable.

Phenomenon	Operator-Count Prediction	Experimental Reality
 The Double-Negation Asymmetry	2x processing cost.	Cost is non-additive. Double-reversal configurations are often verified faster than single negations.
 Domain Sensitivity of Monotonicity	Zero cost for downward-entailing quantifiers without overt 'not' (e.g., 'fewer than half').	High processing cost tracks the boundary structure of the semantic domain, independent of word count.
 Sustained Verification Effects	Cost is a parsing artifact; a temporal delay before verification should eliminate the effect.	Negation effects persist across time delays. The representation itself remains structurally distorted.

The Paradigm Shift: From Truth Values to Geometry

The three experimental failures share a common diagnostic feature. Processing cost tracks the geometry of the inferential environment, not the inventory of lexical items. Negation is not fundamentally a truth-value operator. It is a geometric transformation of the distinction space.

The Old View

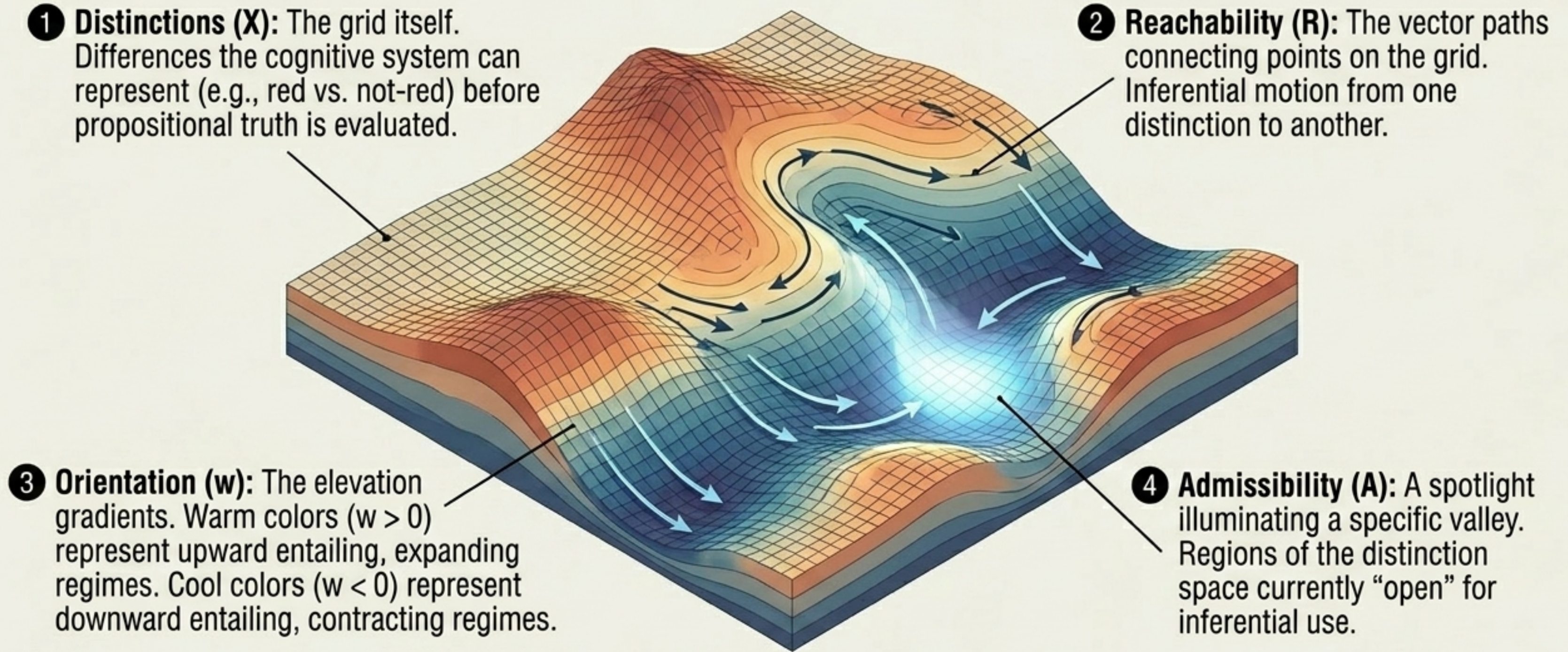


The New View



Monotonicity is a property of the orientation field, not individual words.

Anatomy of an Inferential Field

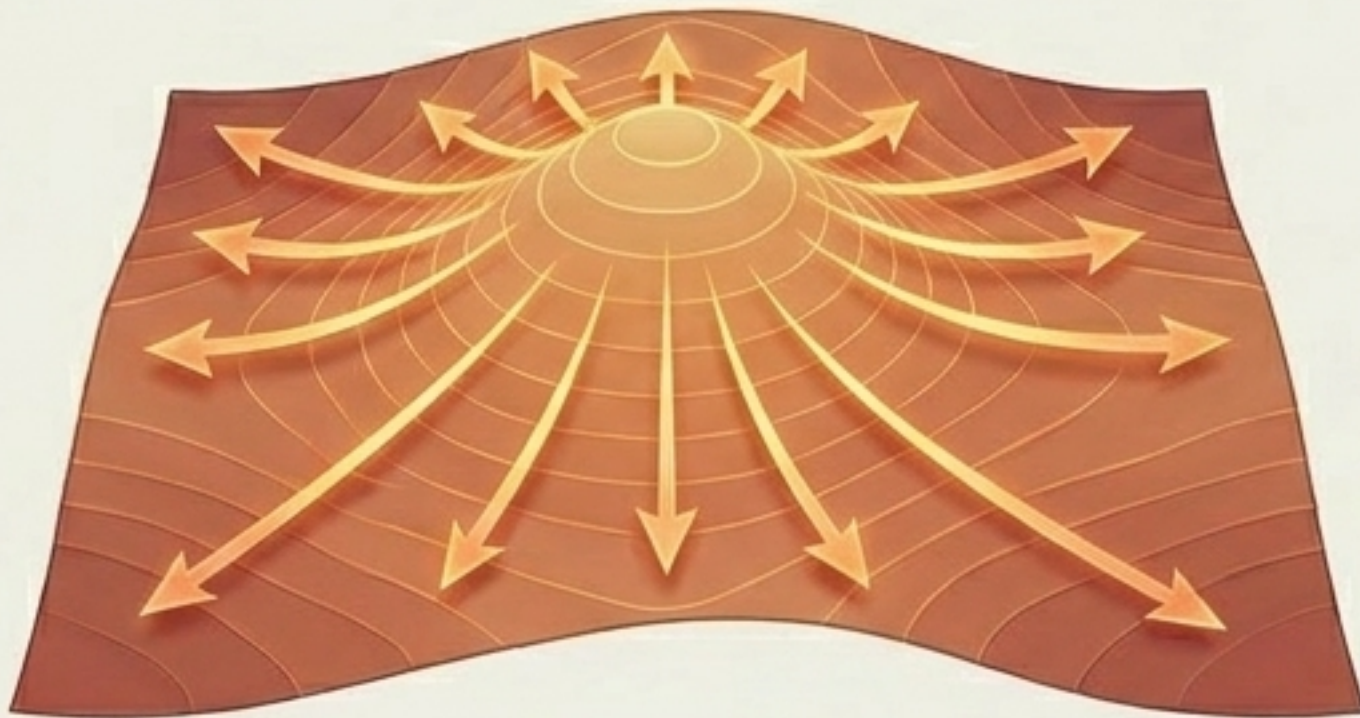


Inferential fields are geometric structures composed of these fundamental elements, guiding cognitive motion.

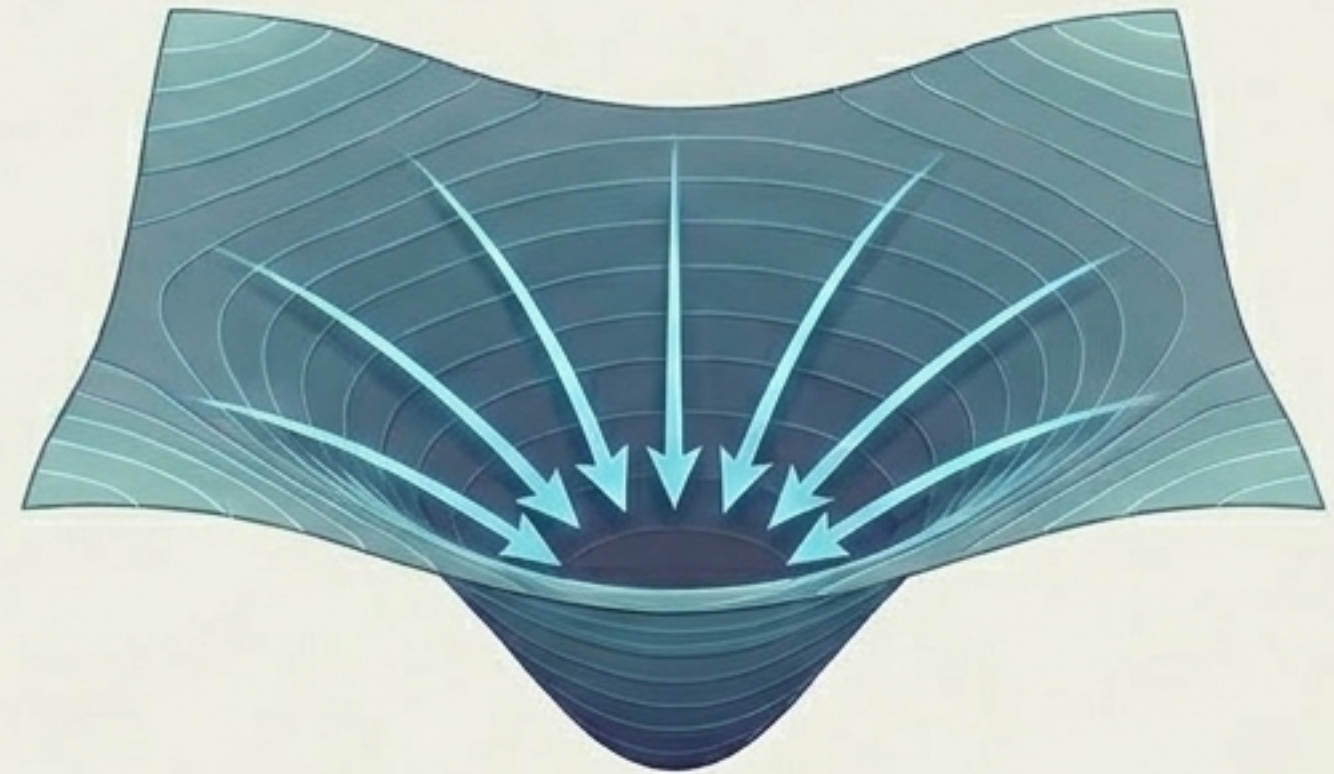
Negation as Orientation Reversal

Negation does not change facts; it reverses the direction of inferential flow. The negation operator N acts on the orientation field by pointwise reversal: $N(w) = -w$.

Before: Positive Orientation



After: Negation Applied

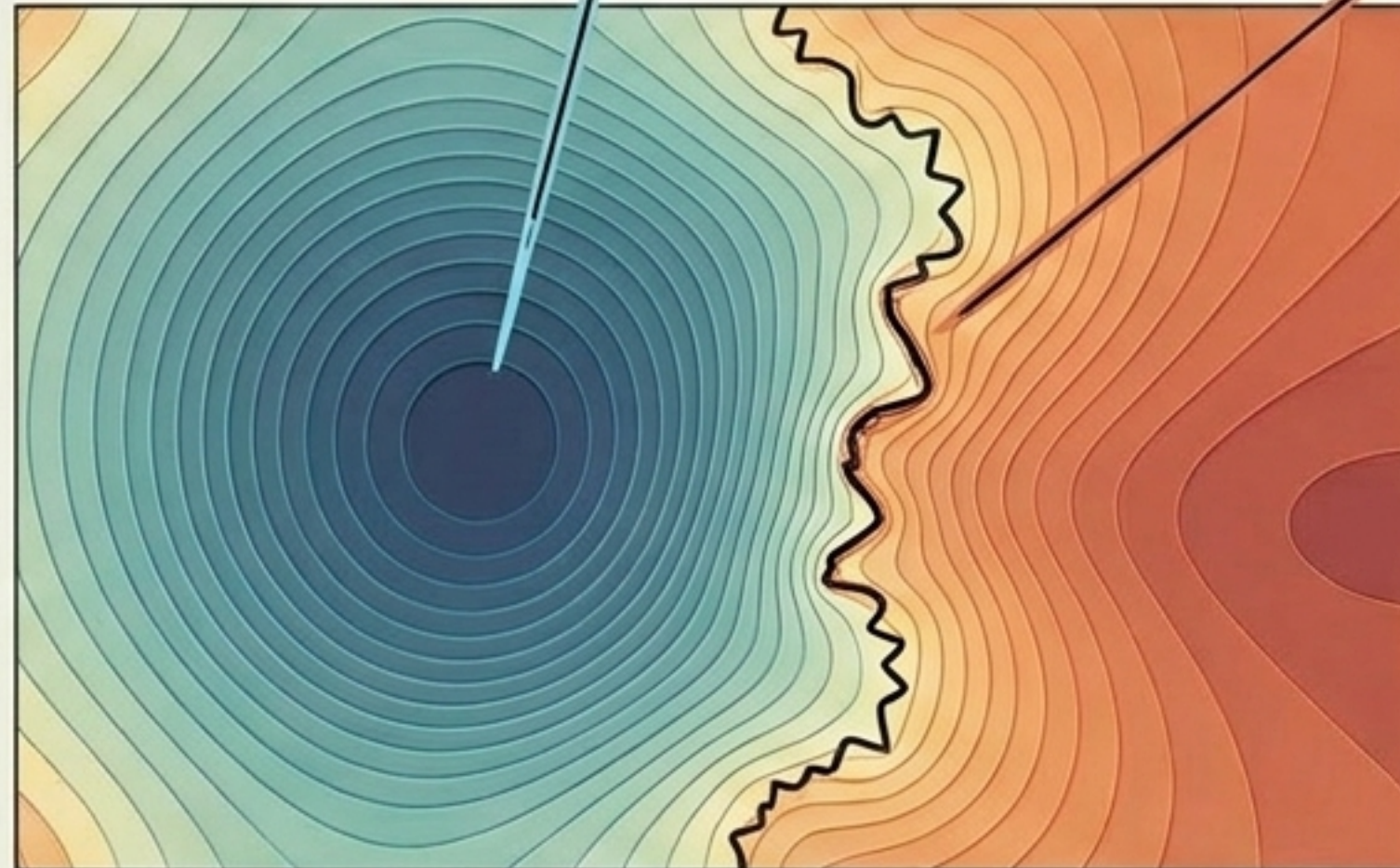


$N(N(w)) = w$. Two structural reversals return the field to its original orientation, explaining why double negation is not additive.

Deconstructing Cognitive Cost

Processing cost is determined by field distortion, not operator count. It isolates distinct quantities: polarity cost, monotonicity cost, and boundary cost.

$$C(x) = a \cdot 1[w(x) < 0] + B \cdot K_w(x)$$

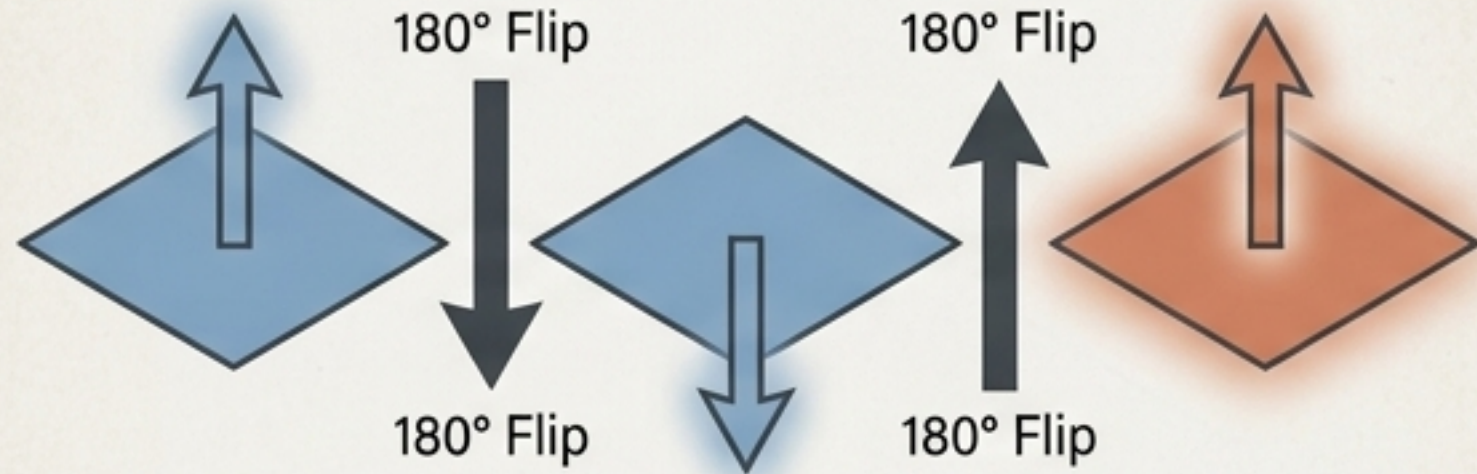


Orientation Cost: The baseline friction of moving through a “downward” (negative) regime.

Boundary Cost: The sharp curvature penalty of crossing between upward and downward domains.

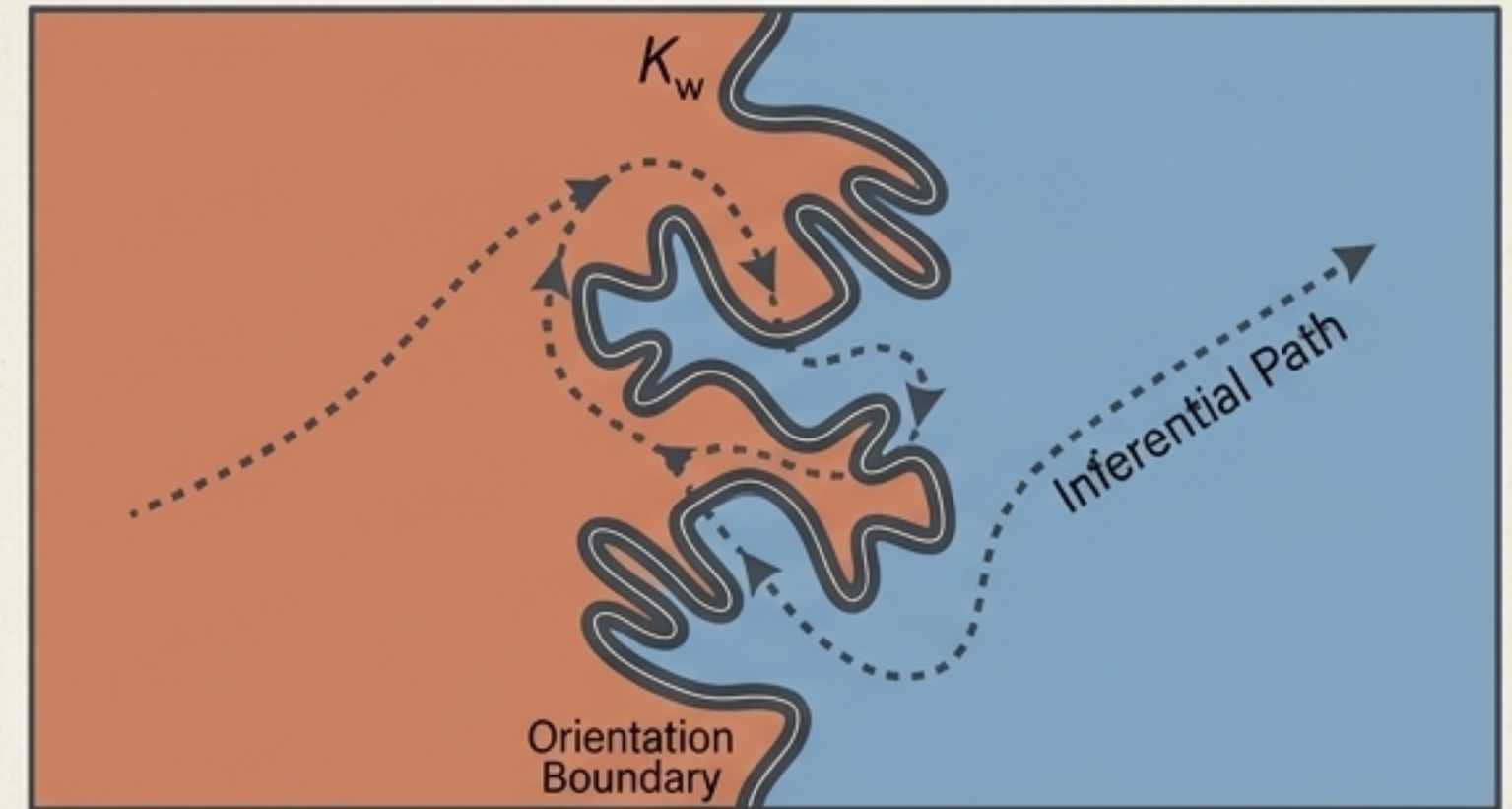
Resolving the Anomalies: Cancellation and Curvature

Double Negation Non-Additivity



Two consecutive flips return the field to its original orientation. The net orientation cost is zero. Cost is bounded by boundary traversal, not multiplied by two.

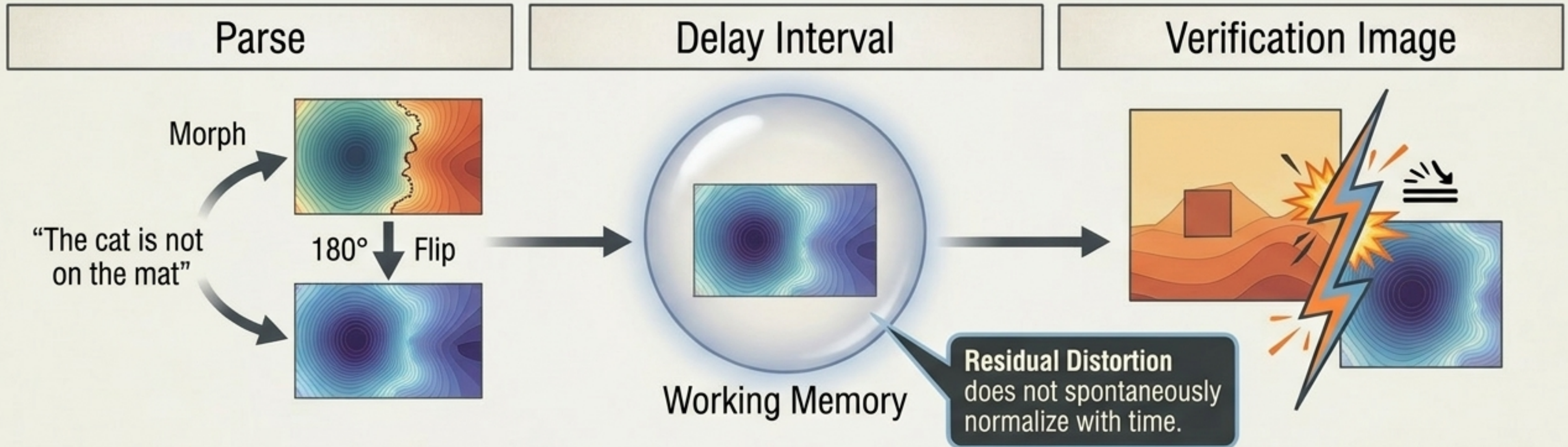
Domain-Sensitive Cost



Cost tracks domain structure. Two configurations with equivalent domain structures produce equivalent costs. The system pays the toll of navigating the curvature of the boundary, regardless of lexical words.

Resolving the Anomalies: Residual Distortion in Working Memory

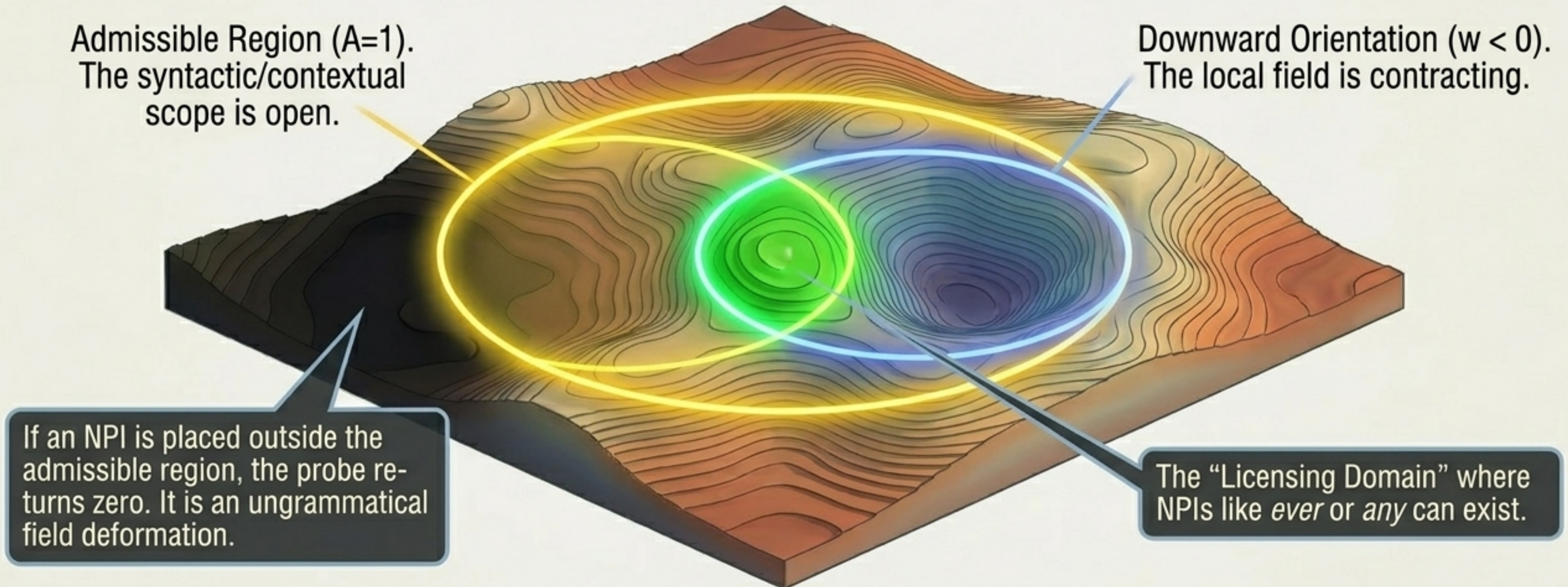
Why does negation cost persist across a time delay? Because working memory stores the distorted topological structure, not just the parsed words.



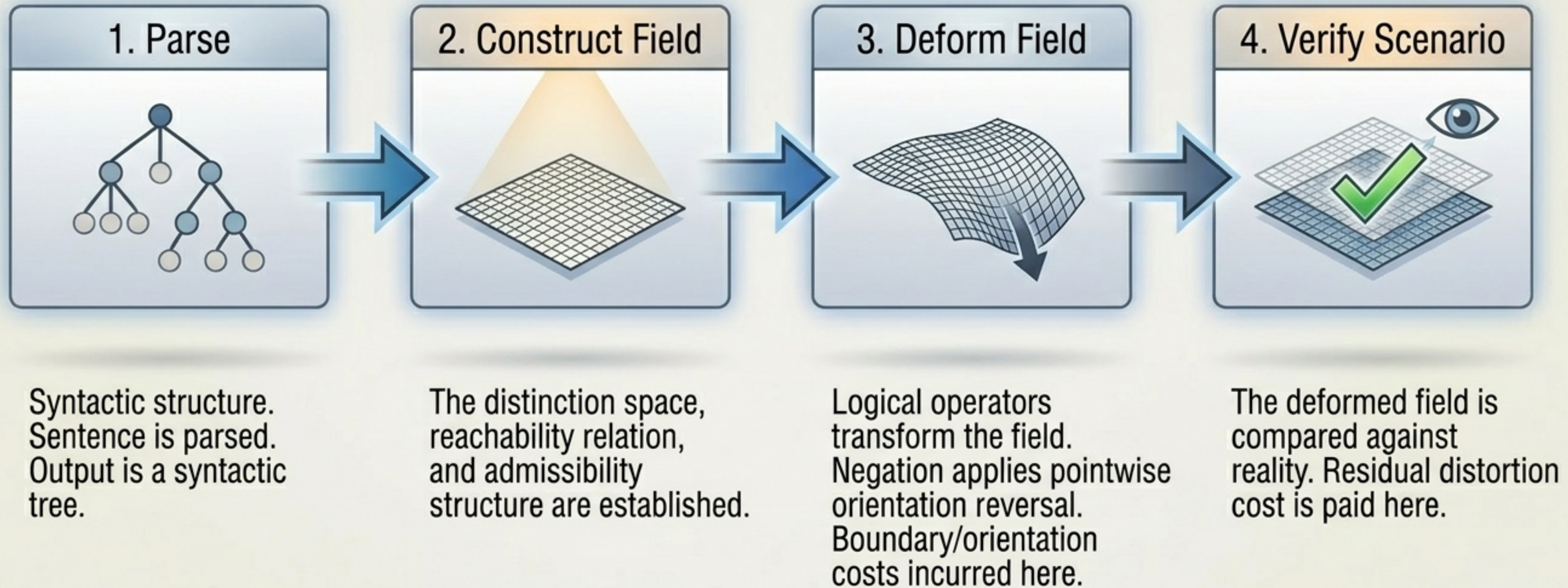
Conclusion: The system must pay the computational toll to actively re-orient the stored field against the default-oriented scenario. Time does not resolve structural distortion.

NPIs as Geometric Probes

Standard accounts state NPIs are triggered by “downward entailment.” The field framework proves NPIs are geographic probes. They require two conditions simultaneously to survive, asserting reachability across regions where contraction usually prevents it.

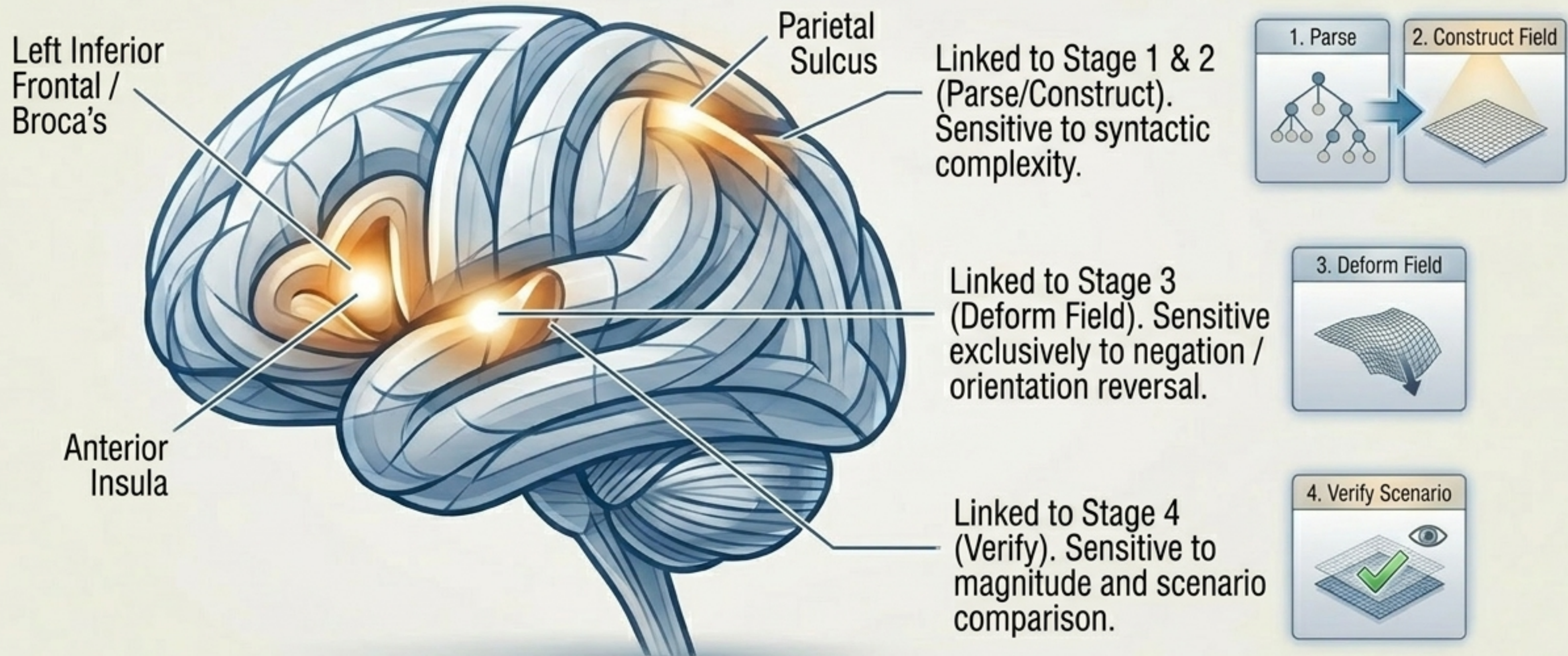


The Four-Stage Functional Architecture



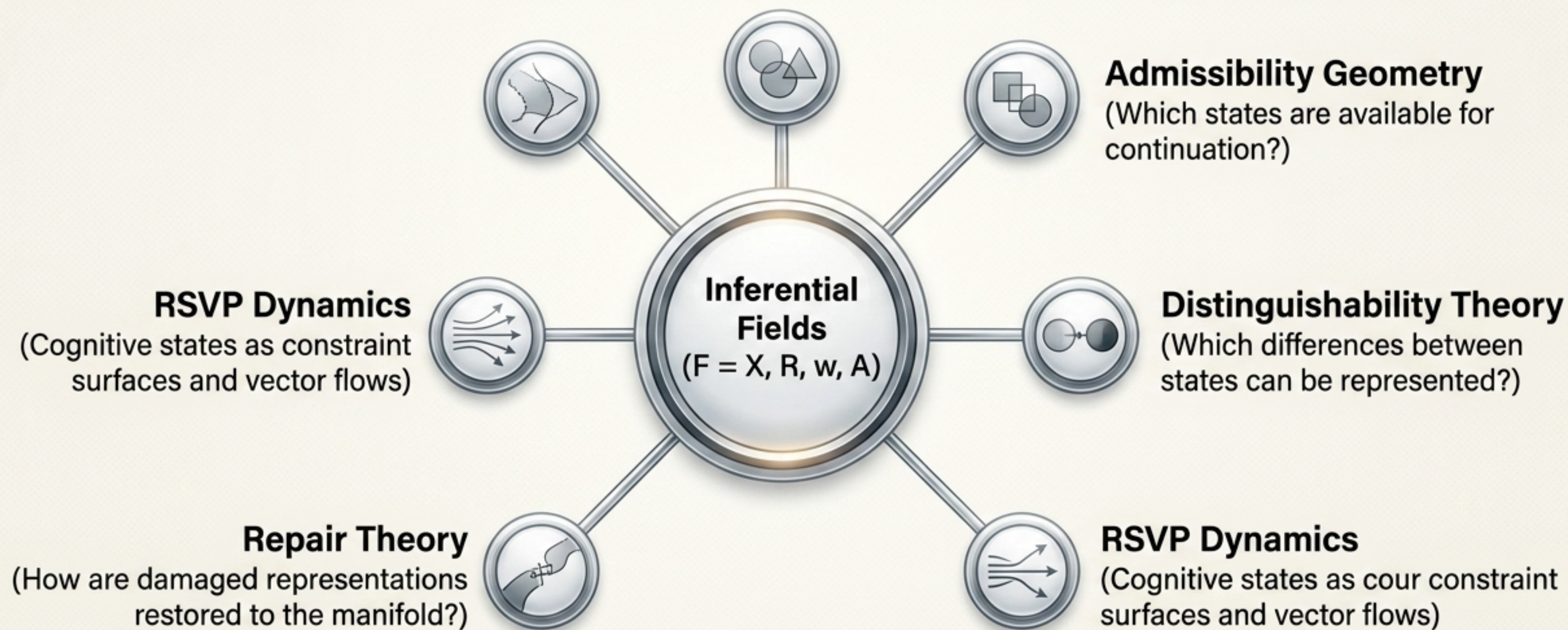
Grounding the Model: Neural Dissociation

The 4-stage model is constrained by empirical fMRI data. A triple neural dissociation proves these are functionally separable neural operations, not just a generic “task difficulty” load.



The Broader Mathematical Unification

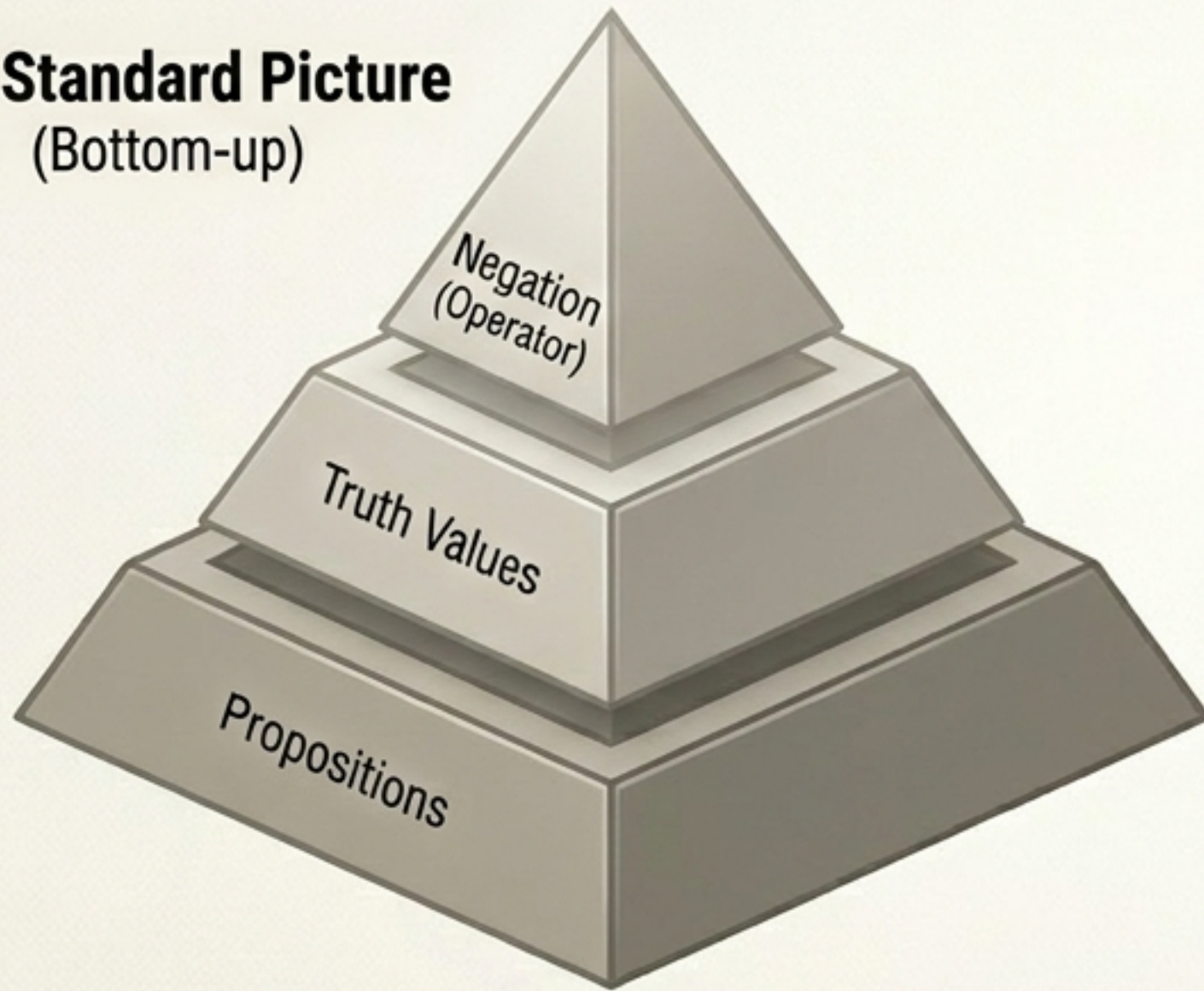
The Inferential Field framework does not exist in isolation. The exact same mathematical structure recurs across independent domains of cognition.



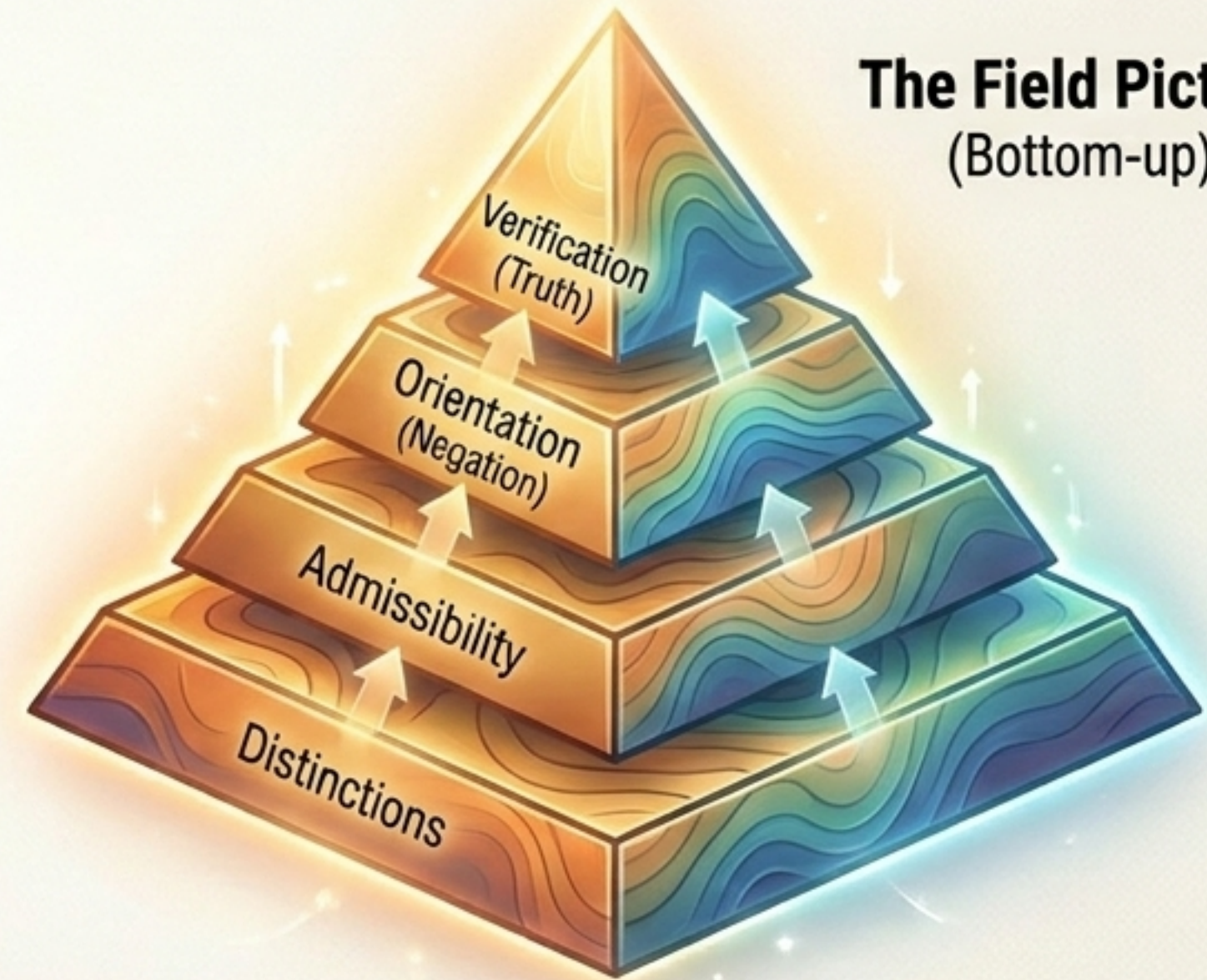
The Philosophical Inversion

Logic does not begin with truth and add negation. It begins with the organization of distinctions. Negation transforms the orientation of those distinctions. Truth is a derived notion.

The Standard Picture
(Bottom-up)



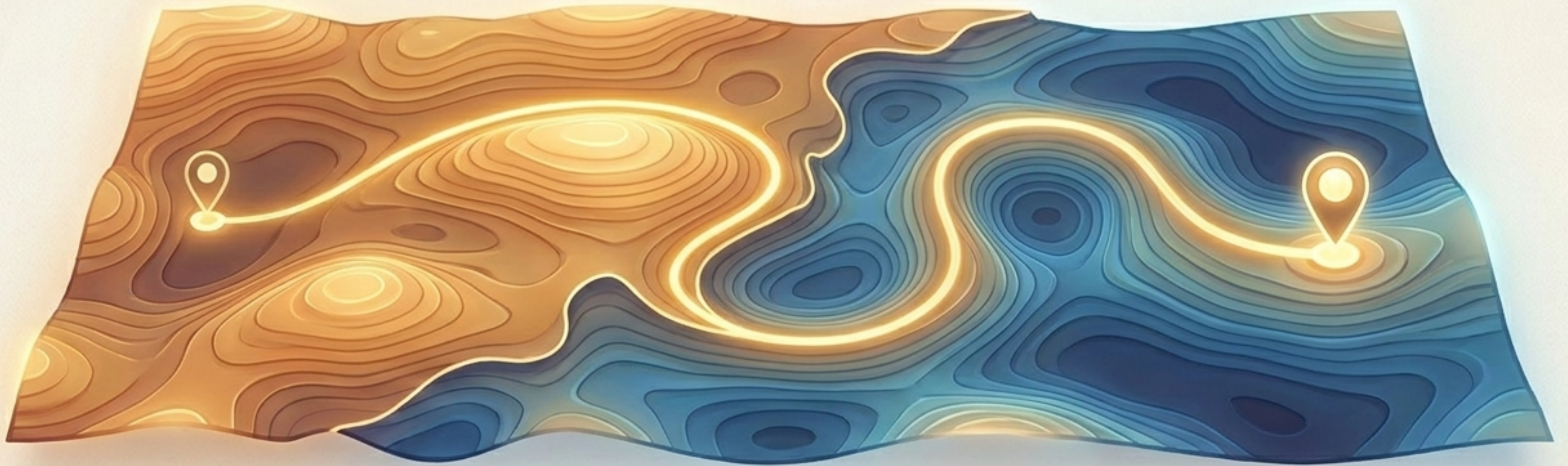
The Field Picture
(Bottom-up)



Negation is defined at the level of orientation structure, independent of and prior to the assignment of truth values.

Truth as Verified Reachability

Negation is not an addition to logic. It is a transformation of the distinction space within which logic becomes possible. Truth is what happens when you successfully navigate an oriented inferential field.



Admissibility -> Orientation -> Truth